Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. All experiments were run on a single GPU with 12 GB of memory.

2 Results

Figure 1 shows the test accuracy of ViT-S by dataset.

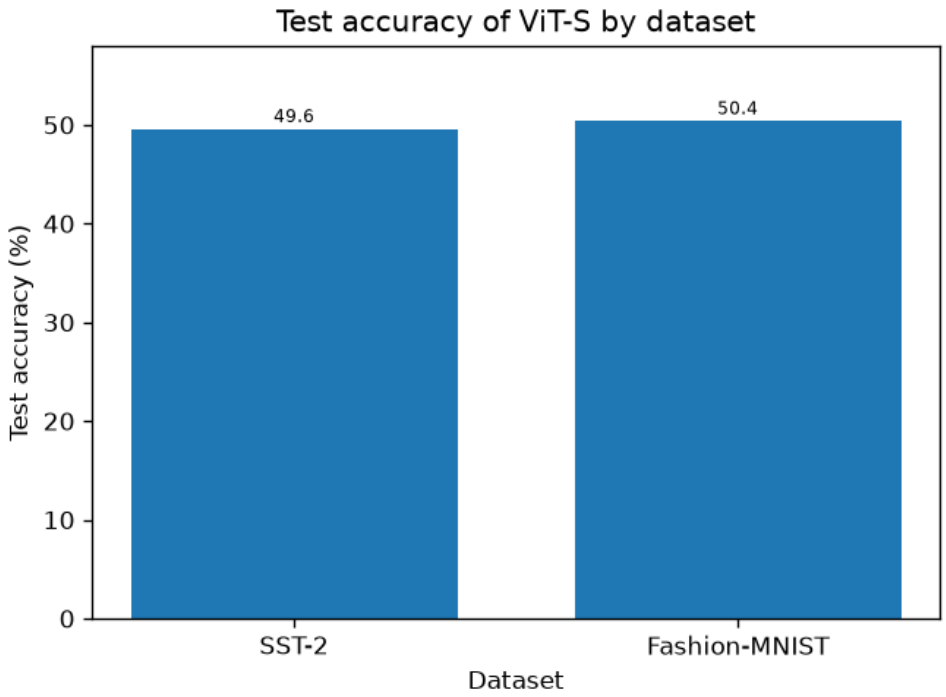

Figure 1: Test accuracy of ViT-S by dataset for the models in this experiment.

On Fashion-MNIST, ViT-S obtains a test accuracy of 79.4%. On Fashion-MNIST, ViT-S obtains a test accuracy of about 50%. ViT-S reaches 53.3% test accuracy on Fashion-MNIST (Figure 1).

Figure 2 shows the validation accuracy of ResNet-50 by dataset.

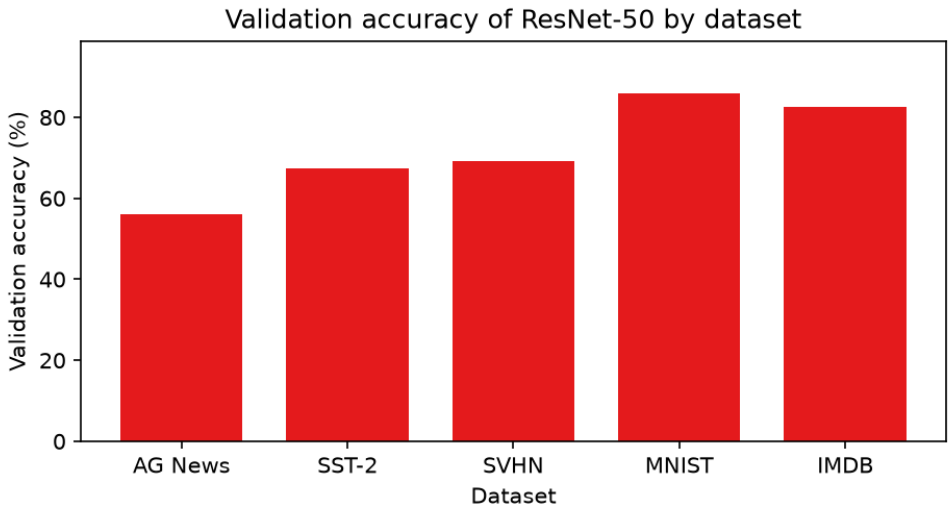

Figure 2: Validation accuracy of ResNet-50 by dataset for the models in this experiment.

ResNet-50 reaches 67.4% validation accuracy on SST-2 (Figure 2). As shown in Figure 2, ResNet-50 achieves 65.4% validation accuracy on SST-2. ResNet-50 achieves 67.4% validation accuracy on SST-2.

3 Discussion

On Tiny-ImageNet, Naive Bayes obtains an accuracy of 84.8%. The differences between the models are discussed in the next section.